

Ex 3.1

① Let $A = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$

ii) $A - B = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} - \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$

$= \begin{bmatrix} 2+1 & -3-3 & -5-5 \\ -1-1 & 4+3 & 5+5 \\ 1+1 & -3-3 & -4-5 \end{bmatrix}$

$A - B = \begin{bmatrix} 3 & -6 & -10 \\ -2 & 7 & 10 \\ 2 & -6 & -9 \end{bmatrix}$

iii) $2A + 3B = 2 \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} + 3 \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$

$= \begin{bmatrix} 4 & -6 & -10 \\ -2 & 8 & 10 \\ 2 & -6 & -8 \end{bmatrix} + \begin{bmatrix} -3 & 9 & 15 \\ 3 & -9 & -15 \\ -3 & 9 & 15 \end{bmatrix}$

$= \begin{bmatrix} 4-3 & -6+9 & -10+15 \\ -2+3 & 8-9 & 10-15 \\ 2-3 & -6+9 & -8+15 \end{bmatrix}$

$= \begin{bmatrix} 1 & 3 & 5 \\ 1 & -1 & -5 \\ -1 & 3 & 7 \end{bmatrix}$

(v) $AB = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$

$= \begin{bmatrix} -2-3+5 & 6+9-15 & 10+15-25 \\ 1+4-5 & -3-12+15 & -5+20+25 \\ -1-3+4 & 3+9-12 & 5+15-20 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$Q. (ii) \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \begin{bmatrix} ax+hy+gz & hx+by+fz & gx+fy+cz \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \{x(ax+hy+gz) + y(hx+by+fz) + z(gx+fy+cz)\}$$

(iv)

$$\begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^3 = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^2 \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+8 & 2-6 \\ 4-12 & 8+9 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 9-16 & 18+12 \\ -8+68 & -16-51 \end{bmatrix} = \begin{bmatrix} -7 & 30 \\ 60 & -67 \end{bmatrix}$$

$$(v) \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^3$$

$$= \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^2 \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0+0 & -1-1+0 & -1+0-1 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 0+0+0 & 0+0+0 & 0+0+1 \end{bmatrix} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{pmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{pmatrix}$$

$$B = \begin{pmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{pmatrix}$$

$$AB = \begin{pmatrix} \cos^2 \theta \cos^2 \phi + \cos \theta \sin \theta \cos \phi \sin \phi & \cos^2 \theta \cos \phi \sin \phi + \cos \theta \sin \theta \sin^2 \phi \\ \cos \theta \sin \theta \cos^2 \phi + \sin^2 \theta \cos \phi \sin \phi & \cos \theta \sin \theta \cos \phi \sin \phi + \sin^2 \theta \sin^2 \phi \end{pmatrix}$$

$$= \begin{pmatrix} \cos \theta \cos \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) & \cos \theta \sin \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) \\ \sin \theta \cos \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) & \sin \theta \sin \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) \end{pmatrix}$$

$$= \begin{pmatrix} \cos \theta \cos \phi \cos(\theta - \phi) & \cos \theta \sin \phi \cos(\theta - \phi) \\ \sin \theta \cos \phi \cos(\theta - \phi) & \sin \theta \sin \phi \cos(\theta - \phi) \end{pmatrix}$$

$$= \cos(\theta - \phi) \begin{pmatrix} \cos \theta \cos \phi & \cos \theta \sin \phi \\ \sin \theta \cos \phi & \sin \theta \sin \phi \end{pmatrix}$$

(given) $\theta - \phi = K \frac{\pi}{2}$ where K is odd $K = 1, 3, 5, 7, \dots$

$$\therefore AB = \cos\left(K \frac{\pi}{2}\right) \begin{pmatrix} \cos \theta \cos \phi & \cos \theta \sin \phi \\ \sin \theta \cos \phi & \sin \theta \sin \phi \end{pmatrix}$$

$$= 0 \cdot \begin{pmatrix} \cos \theta \cos \phi & \cos \theta \sin \phi \\ \sin \theta \cos \phi & \sin \theta \sin \phi \end{pmatrix}$$

$$= \mathbf{O} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$\therefore \cos K \frac{\pi}{2} = 0$ where K is odd.

$$\textcircled{4} \begin{pmatrix} \lambda_1^2 & \lambda_1 \lambda_2 & \lambda_1 \lambda_3 \\ \lambda_1 \lambda_2 & \lambda_2^2 & \lambda_2 \lambda_3 \\ \lambda_1 \lambda_3 & \lambda_2 \lambda_3 & \lambda_3^2 \end{pmatrix} \begin{pmatrix} M_1^2 & M_1 M_2 & M_1 M_3 \\ M_1 M_2 & M_2^2 & M_2 M_3 \\ M_1 M_3 & M_2 M_3 & M_3^2 \end{pmatrix}$$

$$\lambda_1^2 M_1^2 + \lambda_1 \lambda_2 M_1 M_2 + \lambda_1 \lambda_3 M_1 M_3$$

$$\lambda_1^2 M_1 M_2 + \lambda_1 \lambda_2 M_2^2 + \lambda_1 \lambda_3 M_2 M_3$$

$$\lambda_1^2 M_1 M_3 + \lambda_1 \lambda_2 M_2 M_3 + \lambda_1 \lambda_3 M_3^2$$

$$\lambda_1 \lambda_2 M_1^2 + \lambda_2^2 M_1 M_2 + \lambda_2 \lambda_3 M_1 M_3$$

$$\lambda_1 \lambda_2 M_1 M_2 + \lambda_2^2 M_2^2 + \lambda_2 \lambda_3 M_2 M_3$$

$$\lambda_1 \lambda_2 M_1 M_3 + \lambda_2^2 M_2 M_3 + \lambda_2 \lambda_3 M_3^2$$

$$\lambda_1 \lambda_3 M_1^2 + \lambda_1 \lambda_2 M_1 M_2 + \lambda_3^2 M_1 M_3$$

$$\lambda_1 \lambda_3 M_1 M_2 + \lambda_2 \lambda_3 M_2^2 + \lambda_3^2 M_2 M_3$$

$$\lambda_1 \lambda_3 M_1 M_3 + \lambda_2 \lambda_3 M_2 M_3 + \lambda_3^2 M_3^2$$

$$= \begin{pmatrix} \lambda_1 M_1 (\lambda_1 M_1 + \lambda_2 M_2 + \lambda_3 M_3) & \lambda_1 M_2 (\lambda_1 M_1 + \lambda_2 M_2 + \lambda_3 M_3) & \lambda_1 M_3 (\lambda_1 M_1 + \lambda_2 M_2 + \lambda_3 M_3) \\ \lambda_2 M_1 (\lambda_1 M_1 + \lambda_2 M_2 + \lambda_3 M_3) & \lambda_2 M_2 (\lambda_1 M_1 + \lambda_2 M_2 + \lambda_3 M_3) & \lambda_2 M_3 (\lambda_1 M_1 + \lambda_2 M_2 + \lambda_3 M_3) \\ \lambda_3 M_1 (\lambda_1 M_1 + \lambda_2 M_2 + \lambda_3 M_3) & \lambda_3 M_2 (\lambda_1 M_1 + \lambda_2 M_2 + \lambda_3 M_3) & \lambda_3 M_3 (\lambda_1 M_1 + \lambda_2 M_2 + \lambda_3 M_3) \end{pmatrix}$$

= 0

only when $\lambda_1 M_1 + \lambda_2 M_2 + \lambda_3 M_3 = 0$ i.e. when lines are $\perp$ to each other.

x-----x

⑤

$$(A+B)^2 = (A+B)(A+B) \\ = A^2 + AB + BA + B^2$$

$$(A+B)^2 \neq A^2 + 2AB + B^2 \quad \therefore AB \neq BA$$

Also

$$(A-B)(A+B)$$

$$= A^2 + AB - BA - B^2$$

$$\neq A^2 - B^2$$

$$\therefore AB \neq BA$$

Equality will hold only if $AB = BA$ i.e. A & B commute

x-----x

⑥

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$$

$$A^2 - 4A - 5I = 0$$

$$\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} - 4 \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} - 5 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 0$$

$$\begin{pmatrix} 1+4+4 & 2+2+4 & 2+4+2 \\ 2+2+4 & 4+1+4 & 4+2+2 \\ 2+4+2 & 4+2+2 & 4+4+1 \end{pmatrix} - \begin{pmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{pmatrix} - \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} = 0$$

$$\begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix} - \begin{pmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{pmatrix} - \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} = 0$$

$$\begin{pmatrix} 9-4-5 & 8-8-0 & 8-8-0 \\ 8-8-0 & 9-4-5 & 8-8-0 \\ 8-8-0 & 8-8-0 & 9-4-5 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

Some Special Types of Square Matrix.

(i) Periodic Matrix

A square matrix A is said to be a periodic matrix of period K , if $A^{K+1} = A$

(ii) Idempotent Matrix

$$A^2 = A$$

(iii) Nilpotent Matrix

$$A^P = 0$$

P is index

(iv) Involutory Matrix

$$A^2 = I$$

(v) Symmetric Matrix

$$A^T = A$$

(vi) Skew Symmetric Matrix

$$A^T = -A$$

(vii) Hermitian Matrix

$$(A^T)^* = A$$

(viii) Skew Hermitian Matrix

$$(A^T)^* = -A$$

(ix)

Let A be a matrix over $\mathbb{C}$ (complex numbers) if the elements of A are replaced by their complex conjugate, the resulting matrix is called Conjugate of A and denoted by $\bar{A}$

$$A = \begin{bmatrix} 2+3i & -2 \\ 4 & 5+3i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 2-3i & 2 \\ 4 & 5-3i \end{bmatrix}$$

Some Results

- (i) If A is Symmetric then $A+A^T$ is Symmetric & $A-A^T$ is Skew Symmetric
- (ii) Every square matrix can be written as a sum of Symmetric & skew symmetric matrix

$$A = \frac{1}{2}(A+A^T) + \frac{1}{2}(A-A^T)$$
- (iii) If A is square matrix over $\mathbb{C}$ then $A+(\bar{A})^T$ is Hermitian & $A-(\bar{A})^T$ is Skew Hermitian
- (iv) Every square matrix can be written as a sum of Hermitian & skew Hermitian matrix

$$A = \frac{1}{2}[A+(\bar{A})^T] + \frac{1}{2}[A-(\bar{A})^T]$$

⑦

$$A = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$$

$$A^{K+1} = A$$

K is period

$$A^2 = A \cdot A = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+6-12 & -2-4 & -6-18+18 \\ -3-6+8 & 6+4 & 18+18-27 \\ 2-6 & -4 & -12+9 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -5+18-12 & 10-12 & 30-54+18 \\ 9-30+18 & -18+20 & -54+90-27 \\ -4+12-6 & 8-8 & 24-36+9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} = A$$

$$A^3 = A$$

$$\text{or } A^{2+1} = A$$

hence period = 2.

⑧

$$A = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$$

Nilpotent $A^p = 0$

$$A^2 = A \cdot A = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix} \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 1+3-4 & -3-9+12 & -4-12+16 \\ -1+3+4 & 3+9-12 & 4+12-16 \\ 1+3-4 & -3-9+12 & -4-12+16 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{pmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{pmatrix}$$

Involutory $A^2 = I$

$$A^2 = A \cdot A = \begin{pmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 0+4-3 & 0-3+3 & 0+4-4 \\ 0-12+12 & 4+9-12 & -4-12+16 \\ -12+12 & 3+9-12 & -3-12+16 \end{pmatrix}$$

$$A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$A^2 = I$ hence Involutory

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$$A = \frac{1}{2}(2A)$$

$$= \frac{1}{2}(A+A)$$

$$= \frac{1}{2}(A+A^t - A^t + A)$$

$$A = \frac{1}{2}(A+A^t) + \frac{1}{2}(A-A^t)$$

$$A = B + C \quad \text{--- (1)}$$

Now we prove that B is symmetric
& C is skew symmetric

$$B = \frac{1}{2}(A+A^t)$$

$$B^t = \frac{1}{2}(A+A^t)^t = \frac{1}{2}(A^t + (A^t)^t) = \frac{1}{2}(A^t + A)$$

$$B^t = \frac{1}{2}(A+A^t) = B$$

So B is symmetric

$$\text{Now } C = \frac{1}{2}(A-A^t)$$

$$C^t = \frac{1}{2}(A-A^t)^t = \frac{1}{2}(A^t - (A^t)^t) = \frac{1}{2}(A^t - A)$$

$$C^t = -\frac{1}{2}(A-A^t) = -C$$

So C is skew symmetric

(11)

$$A = \frac{1}{2}(2A)$$

$$= \frac{1}{2}(A+A)$$

$$= \frac{1}{2}[A + (\bar{A})^t - (A)^t + A]$$

$$= \frac{1}{2}[A + (\bar{A})^t] + \frac{1}{2}[A - (A)^t]$$

$$A = B + D \quad \text{--- (1)}$$

We prove that B is Hermitian & D is skew Hermitian:

$$B = \frac{1}{2}[A + (\bar{A})^t]$$

$$\bar{B} = \frac{1}{2}[\overline{A + (\bar{A})^t}] = \frac{1}{2}[\bar{A} + (\bar{\bar{A}})^t] = \frac{1}{2}[\bar{A} + (A)^t]$$

$$(\bar{B})^t = \frac{1}{2}[\bar{A} + (A)^t]^t = \frac{1}{2}[(\bar{A})^t + (A^t)^t]$$

$$= \frac{1}{2}[(\bar{A})^t + A] = \frac{1}{2}[A + (\bar{A})^t] = B$$

$$(\bar{B})^t = B$$

Hence B is Hermitian.

$$D = \frac{1}{2}[A - (\bar{A})^t]$$

$$\bar{D} = \frac{1}{2}[\overline{A - (\bar{A})^t}] = \frac{1}{2}[\bar{A} - (\bar{\bar{A}})^t] = \frac{1}{2}[\bar{A} - (A)^t]$$

$$(\bar{D})^t = \frac{1}{2}[\bar{A} - (A)^t]^t = \frac{1}{2}[(\bar{A})^t - (A^t)^t] = \frac{1}{2}[(\bar{A})^t - A]$$

$$= -\frac{1}{2}[A - (\bar{A})^t] = -D$$

$$(\bar{D})^t = -D$$

Hence D is skew Hermitian.

(12)

A & B are symmetric Matrices

$$\text{So } A^t = A \text{ --- (1) \& } B^t = B \text{ --- (2)}$$

Suppose A & B commute i.e. AB = BA & Prove $(AB)^t = AB$

$$(AB)^t = B^t A^t = BA = AB$$

Hence AB is symmetric

using (1) & (2)

Supposition

Now suppose $(AB)^t = AB$

• Prove $AB = BA$

$$(AB)^t = AB$$

$$B^t A^t = AB$$

$$BA = AB \quad \text{using ① \& ②}$$

Hence A & B commute.

⑬ Let A is symmetric i.e. $A^t = A$ — ①

We prove $B = P^t A P$ is symmetric

$$B^t = (P^t A P)^t$$

$$= P^t A^t (P^t)^t$$

$$B^t = P^t A P \quad \because A^t = A \text{ \& } (P^t)^t = P$$

$$B^t = B \quad \text{Hence } B \text{ is symmetric}$$

Now let A is skew symmetric i.e. $A^t = -A$ — ②

We prove $B = P^t A P$ is skew symmetric

$$B^t = (P^t A P)^t$$

$$= P^t A^t (P^t)^t$$

$$= P^t (-A) P$$

$$= -P^t A P$$

$$B^t = -B$$

Hence B is skew symmetric.

X-----X

(14) To Prove AA^t & A^tA are symmetric for a square matrix A

Let A be a square matrix

$$\text{Now } (AA^t)^t = (A^t)^t (A)^t$$

$$(AB)^t = B^t A^t$$

So AA^t is symmetric.

$$\text{Now } (A^tA)^t = A^t (A^t)^t$$

$$= A^t A$$

So A^tA is also symmetric.

(15) Let $A^n + a_{n-1}A^{n-1} + \dots + a_1A + a_0I$

Let A is symmetric. So $A^t = A$.

First we prove that $(A^n)^t = (A^t)^n$ where n is the integer

$$\begin{aligned} \text{Now } (A^n)^t &= (A \cdot A \cdot \dots \text{ n times})^t \\ &= A^t \cdot A^t \cdot \dots \text{ n times} \\ &= (A^t)^n \end{aligned}$$

Now we prove $B = A^n + a_{n-1}A^{n-1} + \dots + a_1A + a_0I$ is symmetric

$$\begin{aligned} B^t &= (A^n + a_{n-1}A^{n-1} + \dots + a_1A + a_0I)^t \\ &= (A^n)^t + (a_{n-1}A^{n-1})^t + \dots + (a_1A)^t + (a_0I)^t \\ &= a_n(A^n)^t + a_{n-1}(A^{n-1})^t + \dots + a_1(A)^t + a_0(I)^t \\ &= a_n(A^t)^n + a_{n-1}(A^t)^{n-1} + \dots + a_1(A^t) + a_0I \\ &= a_nA^n + a_{n-1}A^{n-1} + \dots + a_1A + a_0I \end{aligned}$$

$$\therefore B^t = B$$

$$B^t = B$$

Hence $A^n + a_{n-1}A^{n-1} + \dots + a_1A + a_0I$ is symmetric

⑥ Let $B = A + (\bar{A})^t$

$$\bar{B} = \overline{A + (\bar{A})^t} = \bar{A} + (\bar{\bar{A}})^t = \bar{A} + (A)^t$$

$$(\bar{B})^t = [\bar{A} + (A)^t]^t = [(\bar{A})^t + (A^t)^t] = [(\bar{A})^t + A]$$

$$(\bar{B})^t = A + (\bar{A})^t = B$$

Hence $B = A + (\bar{A})^t$ is Hermitian.

Let $C = A(\bar{A})^t$

$$\bar{C} = \overline{A(\bar{A})^t} = \bar{A}(\bar{\bar{A}})^t = \bar{A}(A)^t$$

$$(\bar{C})^t = (\bar{A}A^t)^t = (A^t)^t(\bar{A})^t = (A)(\bar{A})^t$$

$$(\bar{C})^t = C$$

Hence $C = A(\bar{A})^t$ is Hermitian.

Let $D = (\bar{A})^t A$

$$\bar{D} = \overline{(\bar{A})^t A} = (\bar{\bar{A}})^t \bar{A} = (A)^t \bar{A}$$

$$(\bar{D})^t = [(A)^t \bar{A}]^t = (\bar{A})^t (A^t)^t = (\bar{A})^t A$$

$$(\bar{D})^t = D$$

Hence D is Hermitian.

Imp

⑩

Let A be a Square Matrix

$$\text{Now } A = \frac{1}{2}(2A)$$

$$= \frac{1}{2}(A + A)$$

$$= \frac{1}{2}[A + (\bar{A})^t - (\bar{A})^t + A]$$

$$= \frac{1}{2}[A + (\bar{A})^t] + \frac{1}{2}[A - (\bar{A})^t]$$

$$A = \frac{1}{2}[A + (\bar{A})^t] + i \cdot \frac{1}{2i}[A - (\bar{A})^t]$$

$$A = P + iQ$$

Now we prove that P & Q are Hermitian

$$P = \frac{1}{2}[A + \bar{A}^t]$$

$$(\bar{P})^t = \frac{1}{2}[\overline{A + \bar{A}^t}]^t$$

$$= \frac{1}{2}[\bar{A} + (\bar{\bar{A}})^t]^t$$

$$= \frac{1}{2}[\bar{A} + (A)^t]^t = \frac{1}{2}[(\bar{A})^t + (A^t)^t]$$

$$= \frac{1}{2}[(\bar{A})^t + A] = \frac{1}{2}[A + (\bar{A})^t] = P$$

$(\bar{P})^t = P$ Hence P is Hermitian.

Now $Q = \frac{1}{2i}(A - \bar{A}^t)$

$$(\bar{Q})^t = \frac{1}{2i}(\overline{A - \bar{A}^t})^t = -\frac{1}{2i}(\bar{A} - (\bar{\bar{A}})^t)^t$$

$$= -\frac{1}{2i}(\bar{A} - A^t)^t$$

$$(\bar{Q})^t = -\frac{1}{2i}(\bar{A} - A^t)^t = -\frac{1}{2i}[(\bar{A})^t - (A^t)^t]$$

$$= -\frac{1}{2i}[(\bar{A})^t - A] = \frac{1}{2i}[A - (\bar{A})^t] = Q$$

$(\bar{Q})^t = Q$ Hence Q is Hermitian.

Now we prove the uniqueness of P & Q .

Let $A = R + iS$ — ① (where R & S are Hermitian)

Now $(\bar{A})^t = \overline{R + iS}^t = [\bar{R} - i\bar{S}]^t = (\bar{R})^t - i(\bar{S})^t$

$(\bar{A})^t = R - iS$ — ② $\therefore R$ & S are Hermitian.

Adding ① & ②

$$A + (\bar{A})^t = 2R$$

$$\frac{1}{2}[A + (\bar{A})^t] = R$$

$$P = R$$

Subtracting ② from ①

$$2iS = A - (\bar{A})^t$$

$$S = \frac{1}{2i}[A - (\bar{A})^t] = Q$$

Hence $P + iQ$ is unique.

(21)

Easy

$$\begin{aligned} \bar{\bar{A}} &= A \because A \text{ is real} \\ A^t &= A \because A \text{ is symmetric} \\ \bar{B} &= B \because B \text{ is real} \\ B^t &= -B \because B \text{ is skew symmetric} \end{aligned}$$

To Prove $A + iB$ is Hermitian

$$\text{Let } Q = A + iB$$

$$\bar{Q} = \overline{A + iB}$$

$$= \bar{A} - i\bar{B}$$

$$\bar{Q} = A - iB \because \bar{\bar{A}} = A, \bar{\bar{B}} = B$$

$$(\bar{Q})^t = (A - iB)^t$$

$$= A^t - iB^t$$

$$= A - i(-B) \because A^t = A, B^t = -B$$

$$= A + iB$$

$$(\bar{Q})^t = Q$$

2nd Method

Let P is a Hermitian Matrix, $(\bar{P})^t = P$
 then P is a square Matrix over $\mathbb{C}$

$$\therefore P = [\alpha_{ij} + i\beta_{ij}] \text{ where } \alpha_{ij}, \beta_{ij} \in \mathbb{R}$$

$$= [\alpha_{ij}] + i[\beta_{ij}]$$

$$P = A + iB \text{ where } A = [\alpha_{ij}]$$

$$B = [\beta_{ij}]$$

$$\because \bar{\bar{A}} = A, \bar{\bar{B}} = B$$

$$\bar{P} = \overline{A + iB}$$

$$(\bar{P})^t = (\bar{A} + i\bar{B})^t$$

$$= (A + iB)^t$$

$$(\bar{P})^t = A^t + iB^t$$

$$P = A + iB \because (\bar{P})^t = P$$

$$A + iB = A^t + iB^t$$

$$\therefore P = A + iB$$

Equating we get

$$A = A^t$$

$$B = -B^t$$

 $\therefore A$ is symmetric $\therefore B$ is skew symmetric

Hence A is symmetric & Real
 B is skew symmetric & Real.

$$\text{Let } A = [a_{ij} + i\alpha_{ij}]$$

$$B = [b_{ij} + i\beta_{ij}]$$

$$K = p + iq$$

To Prove $\bar{\bar{A}} = A$

$$\bar{A} = [a_{ij} - i\alpha_{ij}]$$

$$\bar{\bar{A}} = [a_{ij} + i\alpha_{ij}]$$

$$\text{So } \bar{\bar{A}} = A$$

(ii) To Prove $\overline{KA} = \bar{K} \bar{A}$

$$KA = [(p+iq)(a_{ij} + i\alpha_{ij})]$$

$$\overline{KA} = \overline{[(p+iq)(a_{ij} + i\alpha_{ij})]} = [\overline{(p+iq)}] \overline{(a_{ij} + i\alpha_{ij})}$$

$$= [(p-iq)(a_{ij} - i\alpha_{ij})] = (p-iq)[a_{ij} - i\alpha_{ij}]$$

$$\overline{KA} = \bar{K} \cdot \bar{A}$$

(iii) $\overline{A+B} = \bar{A} + \bar{B}$ To Prove.

$$\text{Let } A+B = (a_{ij} + i\alpha_{ij}) + (b_{ij} + i\beta_{ij})$$

$$\begin{aligned}\overline{A+B} &= \overline{(a_{ij} + i\alpha_{ij}) + (b_{ij} + i\beta_{ij})} = \overline{(a_{ij} + i\alpha_{ij})} + \overline{(b_{ij} + i\beta_{ij})} \\ &= (a_{ij} - i\alpha_{ij}) + (b_{ij} - i\beta_{ij})\end{aligned}$$

$$i\alpha_1 + i\alpha_2 = i\alpha_2$$

$$\overline{A+B} = \bar{A} + \bar{B}$$

(iv) $A = (a_{ij} + i\alpha_{ij})$

then $\bar{A} = \overline{(a_{ij} + i\alpha_{ij})} = (a_{ij} - i\alpha_{ij})$

$$(\bar{A})^t = (a_{ij} - i\alpha_{ij})^t$$

$$(\bar{A})^t = (a_{ji} - i\alpha_{ji}) \quad \text{--- ①}$$

Now $A^t = (a_{ij} + i\alpha_{ij})^t$

$$A^t = (a_{ji} + i\alpha_{ji})$$

$$(\bar{A}^t) = \overline{(a_{ji} + i\alpha_{ji})}$$

$$(\bar{A}^t) = (a_{ji} - i\alpha_{ji}) \quad \text{--- ②}$$

$$(\bar{A})^t = \overline{(A^t)}$$

x ----- x

(19) If A is a matrix over the field of real numbers and $AA^t = 0$, show that $A = 0$ (23)

Sol Let $A = [a_{ij}]_{m \times n}$, $A^t = [a_{ji}]_{n \times m}$, $AA^t = 0$ (given)

Now i^{th} row of A is $[a_{i1} \ a_{i2} \ a_{i3} \ \dots \ a_{in}]$

j^{th} column of A^t is $\begin{bmatrix} a_{j1} \\ a_{j2} \\ a_{j3} \\ \vdots \\ a_{jn} \end{bmatrix}$

$$(i,j)^{th} \text{ element of } AA^t = a_{i1}a_{j1} + a_{i2}a_{j2} + a_{i3}a_{j3} + \dots + a_{in}a_{jn}$$

$$(i,i)^{th} \text{ element of } AA^t = a_{i1}^2 + a_{i2}^2 + a_{i3}^2 + \dots + a_{in}^2$$

$\therefore$ Put $i=j$ For Diagonal element

$$0 = a_{i1}^2 + a_{i2}^2 + a_{i3}^2 + \dots + a_{in}^2$$

$$\therefore AA^t = 0 \Rightarrow (i,i)^{th} \text{ element} = 0$$

$$\therefore \text{each } a_{i1} = 0, a_{i2} = 0, \dots, a_{in} = 0$$

$$i = 1, 2, 3, \dots, m$$

Hence each element of $A = 0$

$$\text{So } A = [a_{ij}] = 0$$

(20) If A is Matrix over $\mathbb{C}$ and $A(\bar{A})^t = 0$, show that $A = 0 = \bar{A}$

Sol Let $A = [\alpha_{ij} + i\beta_{ij}]_{m \times n}$, $\bar{A} = [\alpha_{ij} - i\beta_{ij}]_{n \times m}$, $(\bar{A})^t = [\alpha_{ji} - i\beta_{ji}]_{m \times n}$

Now i^{th} row of A is $[\alpha_{i1} + i\beta_{i1} \ \alpha_{i2} + i\beta_{i2} \ \dots \ \alpha_{in} + i\beta_{in}]$

j^{th} col of $(\bar{A})^t$ is $\begin{bmatrix} \alpha_{j1} - i\beta_{j1} \\ \alpha_{j2} - i\beta_{j2} \\ \vdots \\ \alpha_{jn} - i\beta_{jn} \end{bmatrix}$

$$(i,j)^{th} \text{ element of } A(\bar{A})^t = (\alpha_{i1} + i\beta_{i1})(\alpha_{j1} - i\beta_{j1}) + (\alpha_{i2} + i\beta_{i2})(\alpha_{j2} - i\beta_{j2}) + \dots + (\alpha_{in} + i\beta_{in})(\alpha_{jn} - i\beta_{jn})$$

$$(i,i)^{th} \text{ element of } A(\bar{A})^t = (\alpha_{i1} + i\beta_{i1})(\alpha_{i1} - i\beta_{i1}) + (\alpha_{i2} + i\beta_{i2})(\alpha_{i2} - i\beta_{i2}) + \dots + (\alpha_{in} + i\beta_{in})(\alpha_{in} - i\beta_{in})$$

$\therefore$ putting $i=j$ for Diagonal elements

$$0 = (\alpha_{i1}^2 + \beta_{i1}^2) + (\alpha_{i2}^2 + \beta_{i2}^2) + \dots + (\alpha_{in}^2 + \beta_{in}^2)$$

$$\therefore A(\bar{A})^t = 0$$

$$\Rightarrow (i,i)^{th} \text{ element} = 0$$

$$\therefore \text{each } \alpha_{i1} = 0 = \alpha_{i2} = \alpha_{i3} = \dots = \alpha_{in}$$

$$i = 1, 2, 3, \dots, m$$

$$\beta_{i1} = 0 = \beta_{i2} = \beta_{i3} = \dots = \beta_{in}$$

i.e. each

$\therefore$ each element of $A = 0$

$$A = [0 + i0] = 0, \text{ and } \bar{A} = [0 - i0] = 0$$

Ruff

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}_{m \times n}$$

$$A^t = \begin{bmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ a_{13} & a_{23} & \dots & a_{m3} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{bmatrix}_{n \times m}$$

Let $i=1, j=2$

i^{th} row of $A = [a_{11} \ a_{12} \ a_{13} \ \dots \ a_{1n}]$

j^{th} col of $A^t = \begin{bmatrix} a_{21} \\ a_{22} \\ a_{23} \\ \vdots \\ a_{2n} \end{bmatrix}$

$$(i,j)^{th} \text{ element of } AA^t = a_{i1}a_{j1} + a_{i2}a_{j2} + \dots + a_{in}a_{jn}$$

$$\therefore a_{i1} = 0, a_{i2} = 0, \dots, a_{in} = 0$$

$$i = 1, 2, 3, \dots, m$$

i.e.

$$\begin{array}{lll} a_{11} = 0 & a_{12} = 0 & \dots a_{1n} = 0 \\ a_{21} = 0 & a_{22} = 0 & \dots a_{2n} = 0 \\ a_{31} = 0 & a_{32} = 0 & \dots a_{3n} = 0 \\ \vdots & \vdots & \vdots \\ a_{m1} = 0 & a_{m2} = 0 & \dots a_{mn} = 0 \end{array}$$

$$\text{Note } AA^t = 0 \Rightarrow AA^t = \text{zero Matrix}$$

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(24)

Q23

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Find AB using indicated partitioning.

Sol: Let $A_{11} = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$ and $B_{11} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$A_{12} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$B_{12} = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$$

$$A_{21} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$B_{21} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A_{22} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix} \quad \text{--- (1)}$$

Now

$$\begin{aligned} A_{11}B_{11} + A_{12}B_{21} &= \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0+0 & 0+0 & 0+0 \\ 0+0 & 0+0 & 0+0 \end{bmatrix} + \begin{bmatrix} 0+0+0 & 0+0+0 & 0+0+0 \\ 0+0+0 & 0+0+0 & 0+0+0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} A_{11}B_{12} + A_{12}B_{22} &= \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2+2 & 1-4 \\ -4+1 & -2-2 \end{bmatrix} + \begin{bmatrix} 0+0+0 & 0+0+0 \\ 0+0+0 & 0+0+0 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -3 & -4 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} A_{21}B_{11} + A_{22}B_{21} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0+0 & 0+0 & 0+0 \\ 0+0 & 0+0 & 0+0 \\ 0+0 & 0+0 & 0+0 \end{bmatrix} + \begin{bmatrix} 0+0+0 & 0+0+0 & 0+0+0 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 0+0+1 & 0+0+0 & 0+0+0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} A_{21}B_{12} + A_{22}B_{22} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0+0 & 0+0 \\ 0+0 & 0+0 \\ 0+0 & 0+0 \end{bmatrix} + \begin{bmatrix} 0+0+0 & 0+0+0 \\ 0+0+0 & 0+0+0 \\ 0+0+0 & 0+0+0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

from (1) $AB = \begin{bmatrix} 0 & 0 & 0 & 4 & -3 \\ 0 & 0 & 0 & -3 & -4 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$

Inverse of a square matrix

Let A be a square matrix of order n . If there

exists a matrix B of same order n s.t

$$AB = I_n = BA$$

then matrix B is called the inverse of A i.e

$$B = A^{-1} \quad \text{so} \quad AA^{-1} = A^{-1}A = I$$

Theorem Inverse of a square matrix if it exists is unique.

Proof Let B & C are two inverses of a square matrix A

So, by def $BA = AB = I$

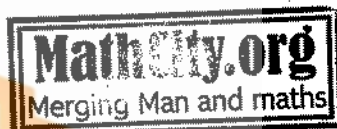
& $CA = AC = I$

Now (Associative Law) $(BA)C = B(AC)$

$$(I)C = B(I)$$

$$C = B$$

Hence inverse is unique.



Q2: $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 6 \\ 0 & 0 & 1 \end{bmatrix}$

Let $A_{11} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $A_{12} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $A_{21} = \begin{bmatrix} 0 & 0 \end{bmatrix}$, $A_{22} = \begin{bmatrix} 2 \end{bmatrix}$
 and $B_{11} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 6 \end{bmatrix}$, $B_{12} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $B_{21} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$, $B_{22} = \begin{bmatrix} 1 \end{bmatrix}$

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}$$

Now $A_{11}B_{11} + A_{12}B_{21} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 6 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 9 & 8 & 15 \\ 19 & 18 & 33 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 15 \\ 19 & 18 & 33 \end{bmatrix}$$

$$A_{11}B_{12} + A_{12}B_{22} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 7 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

$$A_{21}B_{11} + A_{22}B_{21} = \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 6 \end{bmatrix} + \begin{bmatrix} 2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$A_{21}B_{12} + A_{22}B_{22} = \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix} + \begin{bmatrix} 2 \end{bmatrix} = \begin{bmatrix} 2 \end{bmatrix}$$

Hence $AB = \begin{bmatrix} 9 & 8 & 15 & 4 \\ 19 & 18 & 33 & 7 \\ 0 & 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 15 & 4 \\ 19 & 18 & 33 & 7 \\ 0 & 0 & 0 & 2 \end{bmatrix} \text{ Ans.}$

Theorem

Let A & B be non-singular matrices of same order.
 Then AB is non-singular & $(AB)^{-1} = B^{-1}A^{-1}$.

Proof:-

If we show that $(AB)(B^{-1}A^{-1}) = I = (B^{-1}A^{-1})(AB)$
 then obviously AB is non-singular (inverse exists) &
 that its inverse is $B^{-1}A^{-1}$.

$$\begin{aligned} \text{Now } (AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} && \text{Associative Law} \\ &= A(I)A^{-1} \\ &= AA^{-1} \\ &= I && \text{①} \end{aligned}$$

$$\begin{aligned} \text{Also } (B^{-1}A^{-1})(AB) &= B^{-1}(A^{-1}A)B && \text{Associative Law} \\ &= B^{-1}(I)B \\ &= B^{-1}B \\ &= I && \text{②} \end{aligned}$$

So from ① & ② $(AB)^{-1} = B^{-1}A^{-1}$ & AB is non-singular.

Ex 4.2

Q1. Inverse of a Diagonal Matrix is a Diagonal Matrix.

$$\text{Let } D = \begin{bmatrix} d_1 & 0 & 0 & \dots & 0 \\ 0 & d_2 & 0 & \dots & 0 \\ 0 & 0 & d_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_n \end{bmatrix} \quad \& \quad D^{-1} = \begin{bmatrix} d_1^{-1} & 0 & 0 & \dots & 0 \\ 0 & d_2^{-1} & 0 & \dots & 0 \\ 0 & 0 & d_3^{-1} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_n^{-1} \end{bmatrix}$$

If D^{-1} is inverse of D , then we should have

$$DD^{-1} = I_n = D^{-1}D.$$

where I_n is Identity matrix of order n .